

Final Exam Study Guide:

Review all Practices and ICEs. Review Lecture Slides (Week 1 to Week 11)

1. Data Definition Language (DDL): CREATE, ALTER, DROP TABLE, Choosing data types, Defining Constraints (PKs, FKs, CHECK, NOT NULL, UNIQUE)
2. Data Manipulation Language (DML): INSERT, UPDATE SET, DELETE, SELECT FROM WHERE,
3. Data Control Language (DCL): GRANT, REVOKE
4. Transaction Control: COMMIT, ROLLBACK, When changes become permanent
5. Filtering & Aggregation:
 - a. WHERE: Row-level filtering before grouping
 - b. GROUP BY: Generates aggregate sub-tables
 - c. HAVING: Filters after grouping (Use it conditions on COUNT, SUM, etc.)
 - d. COUNT(*), COUNT(col), SUM, AVG, MIN, MAX
 - i. (COUNT(*) counts NULLs; COUNT(col) does not.
6. Set Operators: UNION, UNION ALL, INTERSECT, MINUS
 - a. UNION removes dups; UNION ALL keeps them.
7. Joins & Subqueries
 - a. INNER
 - b. OUTER JOIN: LEFT/RIGHT
 - c. SELECTIN, NOT IN,
 - d. Example:
 - i. Return customers who **do not** have orders via a LEFT JOIN +
WHERE OrderID IS NULL.
 - ii. Rewrite the same logic with a NOT IN subquery.
8. Keys & Integrity
 - a. Primary Key (PK): should be unique, not null, and minimized. Composite PKs are possible when necessary
 - b. Foreign Key (FK): enforces referential integrity (the important concept!!!)
 - c. Alternate (Candidate Key): any other column set that is unique and not null.
 - d. Identifying vs Non-identifying

- i. Identifying: FK is part of child PK (solid line)
 - 1. ORDER & ORDERLINE
- ii. Non-identifying: FK is child's non-key area (dashed line)
 - 1. CUSTOMER & ORDER
- iii. Cardinality (0, 1, and many)

9. Normalization

- a. 1NF: No repeating groups, each field atomic
- b. 2NF: 1NF + no partial dependency on a composite PK
- c. 3NF: 2NF + no transitive dependency on a non-key column
- d. Steps to Normalize
 - i. Identify keys (columns or composites that uniquely determine others)
 - ii. Decompose tables to remove partial/transitive dependencies
 - iii. Validate by reconstructing tables

10. Data Warehousing & Data Mining

- a. ETL
- b. Data mining goals: Explanatory, Confirmatory, Exploratory

11. Enterprise Architecture

- a. Zachman Framework (what is)
- b. Six perspectives (What, Where, How, Why, ..)

12. Integrity & Constraints

- a. Domain (Data types, CHECK (UnitPrice >=0))
- b. Entity (PK uniqueness & NOT NULL)
- c. Referential (FK constraints; Orphan records?)

13. ERD Interpretation

- a. Reading Business Rules -> Entities & Relationships
- b. Find relationships
- c. Resolve many-to-many
- d. Keys & Cardinality
- e. Optionality: Zero-or-one" vs "Exactly-one" on the parent side; controlled by NULL or NOT NULL on the FK.

14. Lab Part Bs with PENGUIN DB & ERD

15. Although the review is comprehensive, it is not exhaustive—be sure to study the lecture slides, practices and ICEs.